

Re-framing Motorsport Data Visualization for Improved Support of Pit Stop Strategy

Jake Ryals

Abstract—This pilot design study explores the development of a new data visualization for the purpose of informing users about the current and predicted near future distribution of vehicles on track in a motor race and its application in making strategic decisions with respect to timing a pit-stop. The project approaches this problem from the perspective that the visualization and data should be re-framed to be relative to user attributes. This article describes how the resultant visualization design is derived from elements of pre-existing visualization methods, feedback from semi-structured interviews and the Five Design-Sheets methodology, and standard data visualization techniques. Additionally, the paper discusses some challenges associated with designing a novel visualization and a proposal for a future evaluation plan.

Index Terms—Motorsport, Data Visualization

1 INTRODUCTION

In the industry of professional motorsport, the ability to make effective strategic decisions is greatly valued. On the topic of strategy in her book on working in motorsport, former Formula 1 strategy engineer Bernie Collins claims, "There are many elements that go into winning a grand prix. Having a competitive and strong car is obviously one of them. Each driver not only needs to be quick but also reliable and capable of sticking to whatever strategy plan has been put in place. Everyone needs to work together as a team. And, ultimately, the strategy must be effective." [4]

In order to have an effective strategy, though, it is necessary that the team has access to the most up to date and relevant information available. In an interview with sports writer Giles Richards, former race engineer Ruth Buscombe recounts, "You might not see a car as a concern but then they have a mega start and that all suddenly changes... You are always reacting to positions, trying to tune your predictions as the race unfolds so you can build a projection of what you think the race will look like and the better projections you have got, the better job you can do. What it comes down to is a big mathletes competition, where you have to do a better job than the other guys." [7] This need to quickly adapt to and interpret new data about the situation of the race is well addressed by the use of data visualizations, which can be designed to handle a stream of live data and be fine tuned to support specific tasks.

This study in particular focuses on advancing data visualization in motorsport specifically for e-sport drivers participating in endurance multiclass races. While the racing environment is still realistic, the participants in these competitions often do not have the same access to resources as established professional motorsport teams. As a result, the visualization tools that these participants have access to may be less robust or lack some features in comparison to their professional counterparts. In particular, I focused on visualization design for assisting users with analyzing situational data for the purpose of making decisions related to pit stop timing due to feedback during semi-structured interviews. (see section 3.1)

1.1 Background

In order to finish longer races, it is necessary for each car to complete some amount of pitstops. While the distances between pit stops are bounded by the need to refuel and change tires, there is typically some margin in how often the driver can perform a pit stop while still completing the optimal number of total pitstops throughout the

race. This margin can be used to potentially gain a strategic advantage. Two common techniques include undercutting and overcutting opponents. Undercutting is described as performing a pitstop earlier than opponents, and overcutting is described as performing a pitstop later than an opponent [5]. By utilizing techniques like these, drivers can overtake opponents without having to fight them on track. However, a driver can only utilize these techniques if they have access to data that demonstrates that the opportunity exists.

These concepts can also be applied to navigating a multi-class race. In endurance sports-car races, there are often multiple classes of cars competing. Typically, lap times vary greatly in between classes. As a result, cars between classes are typically not battling for position, and consequently, it is beneficial for drivers in this case to let the faster car by. However, this process typically slows both cars down somewhat in comparison to driving alone, so situations where different classes interact should be avoided when possible. Since pitstops can facilitate overtakes without fighting, though, drivers can use this to their advantage - if a faster car is approaching a slower car, the slower one can make an early pit stop and prevent the loss of pace that stems from that additional overtake on track. In order to make well informed decisions on how to navigate these situations most effectively, drivers need to know where cars are on track relative to them and where they will be in a few laps time.

2 RELATED WORK

In this section, I discuss some commonly used visualization tools used to inform users of information that is prerequisite to strategic decision making, primarily focusing on their strengths and shortcomings. I will later synthesize these observations in order to inform the design of a visualization that holistically incorporates the utility that these visualizations provide while negating any disadvantageous aspects. These resources were identified via the semi-structured interviews discussed in section 3.1.

2.1 The Circle of Doom [6]

The circle of doom is one of few motorsport visualizations to represent track position in a more abstract manner. Instead of placing marks on a map of the track to represent vehicle position, the track map is instead represented as a circle. This design choice allows the location of marks to instead represent lap progression, allowing for the angular distance between marks to represent the time-based difference between cars on track. (in this context, time-based distance can be interpreted in the following way: if two cars are, for example, three seconds apart, then if the car in front were to come to a complete stop and the following car were to continue as normal, then it would take three seconds for the car behind to pass the car ahead)

However, it is not clear how well humans are capable of judging differences in angles compared to euclidean distance. To assuage this problem, the visualization can instead be represented on a line segment where both end-points represent the same point in space.

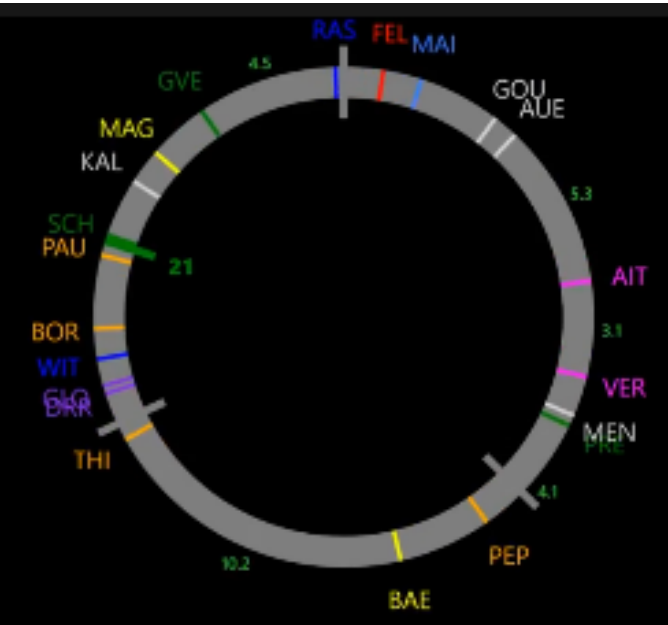


Fig. 1: An example visualization from the Circle of Doom [6]. Cars are plotted as colored lines on the circle, with the longer gray lines representing sector boundaries. Each line is labeled with the driver's initials.

2.2 Emperor Servers Live-timing [1]

As part of its simulation event hosting service, Emperor Servers provides access to a live dashboard containing strategy relevant information such as a live track map and for each car, numeric measures of distances between cars, individual lap times, the number of laps driven, and the amount of and time elapsed since individual pitstops. While this dashboard contains all the information necessary to support strategic analysis, the data is presented in tabular form, making it difficult to parse. Additionally, the track map only encodes vehicle positions. Consequently, in order to synthesize both positional data and other attributes, the user must identify the relevant datum in one visual then locate the corresponding datum in the other visualization. This process is time consuming and thus should be avoided whenever possible.

One additional perspective to consider is that the values displayed in the dashboard are absolute. In the context of strategic decision making, the data analyst typically does not care about these exact values, but rather what they are in comparison to the car that they are doing strategic analysis for. For example, if the user is analyzing lap times using this dashboard, it would be common for them to examine both the competitor's lap time and the user's own time and perform the mental calculation to determine the difference between the two times. If the visualization instead displayed differences in lap times, then the user would not have to complete this task which compromises time and accuracy.

2.3 CMRT [3]

CMRT is a collection of visualizations that are designed to be placed in an overlay on top of the driving simulation, allowing drivers to examine and interpret data from these tools during races. Of the nine visualizations included in this portfolio, those relevant to pitstop strategy are its leaderboard and map.

The leaderboard consists of a table containing information on vehicles' driver name, position, laps completed, distance to the car being driven, and last lap time. The criticisms of Emperor Servers' tabular

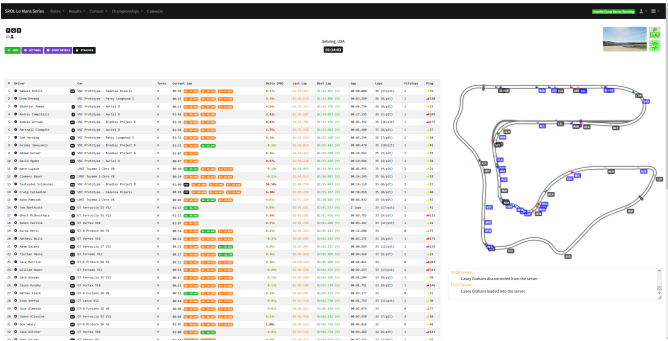


Fig. 2: An example visualization from Emperor Servers [1]. While the marks on the track map use color, this color does not encode any data. To match a mark to the corresponding row in the table, the user must translate driver initials to the full name listed in the table.

and absolute data apply here as well, but CMRT's implementation reveals another drawback to the display of tabular data: due to screen space constraints, the CMRT leaderboard can only display information for six vehicles at a time, rendering the context of the race as a whole inaccessible.

RACE Lap: 2 / 30		GREEN FLAG			
Air: 26° C Tarmac: 24° C		Pos	Int	Last	
1	Yazeed Al-Rajhi	H	L 2	▲ 1 +3.86	3:01.125
3	Thierry Smeets	H	L 2	▲ 3 +0.36	3:04.630
4	Henri Salo	H	L 2	▼ 1 +0.12	3:04.734
5	Player	H	L 2	▲ 7 -	3:04.851
6	Nicolas Gilsoul	H	L 2	▲ 1 -0.95	3:05.863
7	Ott Kapp	H	L 2	▼ 1 -1.47	3:06.335

Fig. 3: The CMRT leaderboard table [3].

The other CMRT component that is relevant to this design study is the map. This visualization plots vehicles on a map in the shape of the track with circles for marks. Each circle has the vehicle's positional ranking encoded in the form of overlaid text. The only other visual encoding that this visualization uses is color: the car that is being focused on and the race leader have unique colors. Otherwise, color encodes whether the car is on a greater lap, the same lap, or a previous lap with respect to the focused car.

2.4 Track Map Display [2]

The track map display application displays positional information similarly to the CMRT map with some additional options for customization. In addition to the color scheme used by CMRT, users have the option to use color to encode which class the vehicle is a member of. This attribute reveals different insights about the vehicle, as cars of different classes typically navigate the track at significantly different speeds. As a result, interactions between cars of different classes are very different than with cars of the same class. Ideally, both attributes would be accessible simultaneously via different visual channels. The designer of the visualization may have chosen to make the availability of these attributes mutually exclusive since they are highly correlated, but there are edge cases where completed laps do not coincide with a specific vehicle class.

Within the customization options of this application, the map can also be represented in the abstract form mentioned at the end of the Circle of Doom section, but this visualization is prone to overdraw.

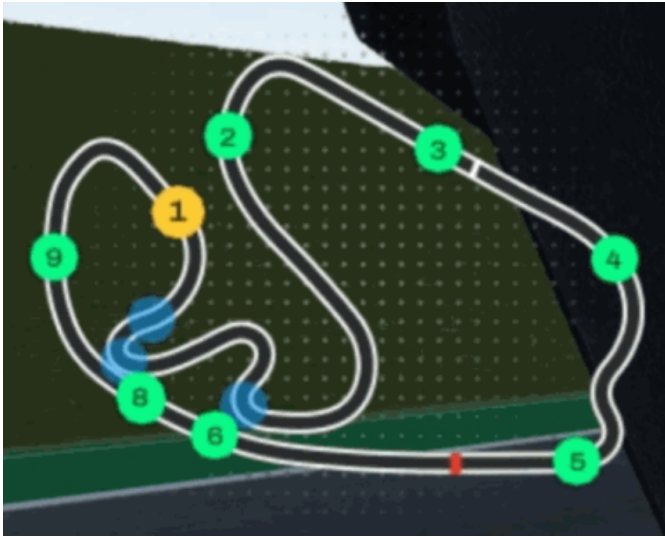


Fig. 4: The CMRT track map [3].

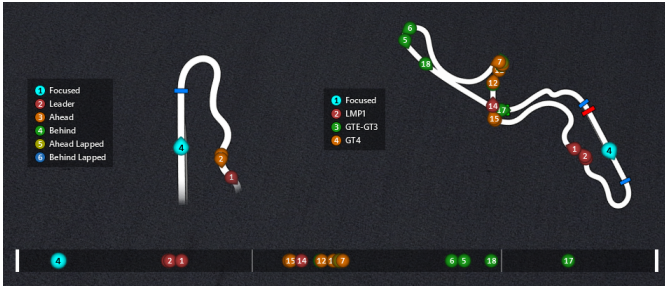


Fig. 5: The visualizations bundled with Track Map [2]. From Track Map's Overtake page: Widgets displayed (left to right): MapDisplay Mini Legend, MapDisplay Mini, MapDisplay Legend, Map Display Full, (second row) MapDisplayProgress.

3 METHODS

3.1 Specification Gathering

In order to determine which strategic tasks were unsupported by the current landscape of motorsport data visualizations, I conducted semi-structured interviews with five e-sport racing drivers. During these interviews, we discussed their holistic workflows for making strategic decisions, including which data visualization tools they used, what they liked and disliked about these visualizations, and which strategic tasks they struggled with or did not bother attempting. From these interviews, it was discovered that there are currently two areas for improvement in most users' workflows: timing pitstops to minimize navigation of traffic from other classes and managing fuel consumption. Because the fuel consumption problem was primarily an issue of statistical modeling and data quality, it was not addressed in this project. Users also expressed interest in a visual design that is compact, not distracting, and able to be quickly glanced at during small periods of time that do not require great cognitive load for driving. Because of these preferences, the visual design has the added constraints of minimizing movement as data is updated in real time in addition to minimizing the need to locate and explore the data.

Following the completion of the semi-structured interviews and literature review, I met again with stakeholders to allow them to participate in the design process using the Five Design Sheets methodology [8] over multiple sessions. During these sessions, we sketched through several low fidelity prototypes and assessed their ability to intuitively reflect the data and fit within the specified constraints.



Fig. 6: The current version of the prototype visualization. Each mark represents one car. The X axis represents the distance to the focused car. The Y axis is used for jittering to prevent overdraw. The different hues represent which relative lap the car is on - green means the car is on the same lap, red means the car is further ahead, and blue means the car is on a previous lap. Opacity encodes how many laps remain on the car's stint - the fewer laps remain, the more transparent the mark is. The shape represents the class of car - in this example, the marks are squares and circles. The "tails" attached to each mark represent the direction and magnitude of the difference in lap times between the car and the focused car. On average, over the course of one lap, the mark will move from where it currently is to the far end of the tail. Any vehicles inside the yellow bar would pass the focused car if it performed a pitstop at the end of the current lap.

3.2 Design Choices

The design choices for the above visualization were made with the intent of mapping the most important features to the most salient visual encodings while also keeping visualization elements intuitive and familiar to users whenever possible. Since positional encodings are the easiest to distinguish, it was the clearest choice for encoding gaps. However, due to the density of marks, it was necessary to let the y axis go unused in favor of jittering the marks. The next best channels, length and orientation, were dedicated to relative difference in lap time. The resulting effect of this choice was that difference in lap times effectively shared the x axis with gaps, which should be intuitive for users since gaps can be interpreted as the sum of the differences in lap times over the course of the race.

The choice to have the spatial range of the visualization be one lap was a result of user feedback - using color for lap differences is a feature of prior work and thus should be familiar to users. This choice also had the effect of reducing the physical footprint of the visualization, making it easier for users to integrate it into their space limited environment (in interviews, most users specified that all of their data visualizations have to fit within a small outer margin of their monitor).

With all those encodings accounted for, that realistically left only lightness/opacity for representing the number of laps remaining until a forced pitstop. The choice to use a transparent rectangular overlay to signify which cars would pass the target upon performing a pitstop can be justified by the fact that a large region is easier to find on the visualization than an alternative, such as a vertical line. This choice also has the effect of highlighting a group of cars that would overtake the target in this situation.

One other important note is that all the data displayed in this visualization (with the exception of laps remaining and class) have been transformed to be relative to the target car instead of absolute. This choice was made to address user feedback about needing to transform the absolute data themselves in order to most effectively interpret it.

4 DISCUSSION

From this pilot study, I discovered the difficulty of visualizing high variety data, the existence of a tradeoff between visualization readability and compactness, and the challenge of introducing new users to a novel visualization.

Given the number of data attributes involved and the constraints at hand, I had to use nearly all available visual encoding channels to create this visualization. This visualization also approaches the limits of human working memory due to the number of attributes displayed. When iterating on this design, these challenges should be kept in mind.

While adjusting the size and aspect ratio of the visualization, I also noticed that the visualization required significantly more height than originally expected. This phenomenon can be explained by the volume of the data and its effect on overdraw. Future work could

include determining recommended minimum visualization heights as a function of the number of data points being shown or applying data summarization techniques to reduce the minimum physical footprint.

While giving early presentations on my work, I also found that directly presenting the prototype to new audiences resulted in a significant amount of confusion and a barrier to understanding. For more unique visualizations like this one, incorporating data storytelling elements to build up the visualization piece by piece may reduce this barrier to understanding.

4.1 Future Evaluation

The next step for this study is to meet again with stakeholders to perform a think-aloud study in order to determine how interpretable the visualization is. To measure this, users would be provided a sample of visualizations with historic race data. After being introduced to the visualization and its encodings, participants would be asked to use the visualization to conclude whether the driver should come in for a pit stop soon or stay out on track. Users would be asked to justify their recommendations in order to ascertain which insights they discovered in the visualization and found most relevant to the task at hand.

REFERENCES

- [1] Assetto corsa server hosting. doi: [game/assetto-corsa](#) 2
- [2] AW939. Track map display (extended) 4.7.1. doi: [downloads/track-map-display-extended.51962/](#) 2, 3
- [3] CMRT Group. Cmrt essential hud 1.0.13. doi: [downloads/cmrt-essential-hud.69475/](#) 2, 3
- [4] B. Collins. *How to Win a Grand Prix*. Quercus Publishing, 2024. 1
- [5] G. Hatton. Optimizing race strategy in motorsports: The power of the undercut and overcut. 1
- [6] Paceteq. Circle of doom. doi: [onetiming/circle-of-doom](#) 1, 2
- [7] G. Richards. *F1 Racing Confidential: Inside Stories from the World of Formula One*. Grand Central Publishing, 2024. 1
- [8] J. C. Roberts, C. Headleand, and P. D. Ritsos. Sketching designs using the five design-sheet methodology. *IEEE Transactions on Visualization and Computer Graphics*, 22(1):419–428, 2016. doi: [10.1109/TVCG.2015.2467271](#) 3